

Loop Groups and Affine Lie Algebras:

G : compact Lie group

$LG \equiv$ Loop Group of G

$$LG \equiv \{ \phi: S^1 \rightarrow G \}$$

$$S^1 \equiv \{ z \in \mathbb{C} \mid |z| = 1 \}$$

• LG is a group under

$$(\gamma_1 \cdot \gamma_2)(z) = \gamma_1(z) \gamma_2(z), \quad \gamma_1, \gamma_2 \in LG$$

(i) $(\gamma_1 \cdot \gamma_2)$ is smooth

(ii) $\gamma_1 \mapsto \gamma_1^{-1}$ is also smooth

LG : ∞ -dim Lie group

* Here we will take $SU(2) = G$

• The loop algebra is the complexified L.A. of $LG \Rightarrow \mathfrak{Lg}^{\mathbb{C}}$

Let $\mathbb{C}[t] \equiv \mathbb{C}$ -algebra of Laurent series

$$f(t) = \sum_{n=-m}^{\infty} C_n t^n$$

for some $m \in \mathbb{Z}$.

• Let $\mathfrak{g}^{\mathbb{C}}$ be the L.A. of G . Then

$$\mathfrak{Lg} = \mathfrak{g} \otimes \mathbb{C}[t]$$

with generators $X \otimes f$, i.e. they are Laurent valued L.A. elements.

$$T \in \mathfrak{Lg} = T^a e^a \quad T^a = T_a^a t^1$$

t^{λ} : basis of $\mathbb{C}[t]$

• This space has a Bracket

$$[X \otimes f, Y \otimes g] \equiv [X, Y] \otimes fg$$

In terms of a basis of generators for each

$$[X^a \otimes t^n, X^b \otimes t^m] = \sum_c i f^{ab}_c X^c \otimes t^{n+m}$$

* This alg. admits a Central extension

$$\hat{\mathfrak{g}} \equiv \mathfrak{Lg} \oplus \mathbb{C}c \quad \mathbb{C}c: 1\text{-dim } \mathbb{C}\text{ vec. space w basis } c$$

* Equip $\hat{\mathfrak{g}}$ w a Lie structure s.t. c belongs to the center of $\mathfrak{Lg}$:

$$[c, \xi] = 0 \quad \forall \xi \in \hat{\mathfrak{g}}$$

* Given a Bilinear form $\omega: \mathfrak{Lg} \times \mathfrak{Lg} \rightarrow \mathbb{C}$ on $\mathfrak{Lg}$ we define a Bracket on $\hat{\mathfrak{g}}$:

$$[\xi + \alpha c, \eta + \beta c] \equiv [\xi, \eta] + \omega(\xi, \eta)c$$

In order that this defines a L.A. it must be that

$$i) \omega(\xi, \eta) = -\omega(\eta, \xi)$$

$$ii) \omega([\xi, \eta], \zeta) + \omega([\eta, \zeta], \xi) + \omega([\zeta, \xi], \eta) = 0$$

* In coordinates

$$\left\{ \begin{array}{l} A \in \mathfrak{g} \otimes \Omega^1(M) \\ A = A^m dx^m = A^m_a (T^a) \otimes dx^a \end{array} \right\}$$

$$[X^a \otimes t^m, X^b \otimes t^n] = \sum_c f^{ab}_c X^c \otimes t^{m+n} + \hat{\kappa} c K(X^a, X^b) v$$

K : Killing form on $\mathfrak{g}$.

* If we expand ξ and η as

$$\xi = X \otimes f, \quad \eta = Y \otimes g$$

$$[X \otimes f + \alpha c, Y \otimes g + \beta c] = [X \otimes f, Y \otimes g] + \omega(X \otimes f, Y \otimes g)c$$

The Jacobi for ω is called the 2-cocycle condition. We can define an equivalence class on the space of 2-forms on $\mathfrak{Lg}$: i.e.

the cohomology $H^2(L\mathfrak{g}, \mathbb{C})$
 * Let $\mu: L\mathfrak{g} \rightarrow \mathbb{C}$ be an 'interior product'

$$\boxed{w \sim w' \Leftrightarrow w(x, y) = w(x, y) + \mu([x, y])}$$

$x, y \in L\mathfrak{g}$

* $\hat{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}$ is the central extension of $L\mathfrak{g}$

Let $\mathfrak{A}$ be a L.A. and M be a left $\mathfrak{A}$ module
 then, the p^{th} cochain group of $\mathfrak{A}$ with coeff. in M is defined as

$$\boxed{C^p(\mathfrak{A}, M) \equiv \text{Hom}_{\mathbb{C}}(\wedge^p \mathfrak{A}, M)}$$

• The differential $d_p: C^p(\mathfrak{A}, M) \rightarrow C^{p+1}(\mathfrak{A}, M)$
 is defined in the usual way yields the
 cohomology group

$$\boxed{H^p(\mathfrak{A}, M) = \ker(d_p) / \text{Im}(d_{p-1})}$$

p^{th} cohomology of the L.A. $\mathfrak{A}$ w, coeff. in M

* Given any element of $H^2(L\mathfrak{g}, \mathbb{C})$ we can
 associate an isomorphism class of the central
 extension of the loop alg. $L\mathfrak{g}$.

$\leadsto \mathbb{C}$: trivial $\mathfrak{g}$ module.

$$\mathfrak{g} \times \mathbb{C} \rightarrow \mathbb{C}$$

* Suppose that

$$0 \rightarrow \mathbb{C} \rightarrow \tilde{L\mathfrak{g}} \xrightarrow{\pi} L\mathfrak{g} \rightarrow 0$$

is the central extension of $L\mathfrak{g}$

• Here $\tilde{L\mathfrak{g}}$ is Bundle of total space $\tilde{L\mathfrak{g}}$
 and base space $L\mathfrak{g}$

$\leadsto$ Take a section $S: L\mathfrak{g} \rightarrow \tilde{L\mathfrak{g}}$

$$\begin{array}{c} \tilde{L\mathfrak{g}} \\ \downarrow \pi \end{array}$$

For $z \in Lg \cong g \otimes \mathbb{C}[t]$, then

$$S(z) \equiv \tilde{z}$$

and set $\Omega(x, y) = [\tilde{x}, \tilde{y}] - \overbrace{[x, y]}^{Lg}$, $x, y \in Lg$
 $\Omega(x, y) = [S(x), S(y)] - S([x, y])$

Claim: Ω determines a cohomology class in $H^2(Lg, \mathbb{C})$ which is independent of S .

$$\Omega: Lg \times Lg \rightarrow \tilde{Lg}$$

$$\hookrightarrow \tilde{Lg} \rightarrow Lg$$

would have to be valued in $\tilde{Lg}$

Claim: there is a one-to-one correspondence between the set of isomorphism classes of a central extension of Lg and $H^2(Lg, \mathbb{C})$

$$\langle \cdot, \cdot \rangle: g \times g \rightarrow \mathbb{C} \quad (\text{Cartan-Killing form})$$

$$\langle [x, y], z \rangle = \langle x, [y, z] \rangle$$

$$\langle x, y \rangle = \text{Tr}(xy), \quad \mathfrak{sl}(2, \mathbb{C})$$

Proposition: For a loop algebra Lg we have $H^2(Lg, \mathbb{C}) \cong \mathbb{C}$. The cohomology $H^2(Lg, \mathbb{C})$ has a basis represented by the 2 cocycle ω defined by

$$\omega(x \otimes f, y \otimes g) = \langle x, y \rangle \text{Res}_{t=0} (dfg)$$

where

$$\text{Res}_{t=0} \left(\sum_n c_n t^n dt \right) = c_{-1}$$

For $\text{Diff}(S^1)$, this makes $G = S^1$ and so we are looking at L.A. of smooth vec. fields over S^1 :

$f(z) \frac{d}{dz}$, $f(z)$: polynomial

$$A = \left\{ f(z) \frac{d}{dz} \mid f(z) \in \mathbb{C}[z, z^{-1}] \right\}$$

w Lie bracket

$$[f\partial, g\partial] = (fg' - gf')\partial$$

$\mathcal{A}$ has a basis $L_n = -z^{n+1}\partial$

$$[L_n, L_m] = (n-m)L_{n+m}$$

$$H^2(A, \mathbb{C}) \cong \mathbb{C}$$

and we have a central extension $\mathcal{V} = A \oplus \mathbb{C}$

$$\left[f \frac{d}{dz} + \xi c, g \frac{d}{dz} + \eta c \right] = [f\partial, g\partial] + \alpha(f\partial, g\partial)c$$

$\alpha \in H^2(A, \mathbb{C})$; 2-cocycle